



Figure 2: GTKWave

```
// Instantiate the Unit Under Test (UUT)
ripple_adder_4bit uut (
    .Sum(Sum),
    .Cout(Cout),
    .A(A),
    .B(B),
    .Cin(Cin)
);
initial begin
    // Initialize Inputs
    A = 0;
    B = 0;
    Cin = 0;
    // Wait 100 ns for global reset to finish
    #100;
    // Add stimulus here
    A=4'b0001;B=4'b0000;Cin=1'b0;
    #10 A=4'b1010;B=4'b0011;Cin=1'b0;
    #10 A=4'b1101;B=4'b1010;Cin=1'b1;
end
initial begin
    $dumpfile("adder.vcd");
    $dumpvars;
end
endmodule
```

Now, in the terminal, `cd` to the folder containing `adder.v` and `adder_tb.v`, and then type the following command:

```
$ iverilog -o adder adder.v adder_tb.v
```

```
$ vvp adder
```

This should generate the `adder.vcd` file. Then type the following command:

```
$ gtkwave adder.vcd
```

This will open the GTKWave window and you will get a window similar to what's shown in Figure 2.

Online Verilog simulator

www.iverilog.com is an online Verilog simulator. The website uses the Icarus Verilog engine but it is not affiliated to the Icarus Verilog project. Waveform display options have not been included in [iverilog.com](http://www.iverilog.com) at the time of writing this article. It can only produce console output. The website has a bunch of example programs for beginners to start with. So if someone wants to try out Verilog before installing [iverilog](http://www.iverilog.com) on their systems, www.iverilog.com is a good option.

An online C-to-Verilog converter

<http://www.c-to-verilog.com> is an online platform on which you can convert programs written in C language to Verilog. This is beneficial to those who are well versed in C but are not so sure about Verilog code. The FAQ section of the website has a few examples that demonstrate the features of [c-to-verilog.com](http://www.c-to-verilog.com).

Icarus Verilog is one of the most popular Verilog simulators in the open source domain. There are many other proprietary software. Verilator is another open source simulator that can synthesise Verilog code to C++ or SystemC. It is up to the readers to further explore the world of logic design. 

References

- [1] 'Verilog-HDL: A Guide to Digital Design and Synthesis' by Samir Palnitkar
- [2] <http://www.verilog.com/>
- [3] <http://en.wikipedia.org/wiki/VHDL>
- [4] <http://iverilog.icarus.com/>

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